Dynamic Energy Budget theory

Day 3: From empirical patterns to formal metabolic theory

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Organized by:

The first mechanistic growth model

Pütter's insight (1920)

Growth is a **surface–volume conflict**:

- Food is taken up across surfaces (gain $\propto V^{2/3}$)
- Maintenance is a whole-body cost (loss $\propto V$)
- Growth stops when gains balance losses

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von Bertalanffy's contribution (1938)

He integrated Pütter's equation and showed that length growth follows a simple curve:

$$\frac{\mathrm{d}L}{\mathrm{d}t} = \dot{r}_B(L_\infty - L)$$

where the growth rate $\dot{r}_B$ and asymptotic length L_∞ summarise the underlying mechanisms.

Kearney (2021)

What Pütter left unresolved

- Pütter could not explain **why** the anabolic coefficient varies with food — he described it as "nutritive material" without a dynamical model
- His theory had **no life stages**: embryos, juveniles, and adults were all the same
- **Reproduction** was not included
- **Respiration** and **stoichiometry** were not addressed

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DEB theory (Kooijman 1986) fills these gaps

It keeps Pütter's surface–volume principle but adds:

- An explicit **reserve** pool that buffers food fluctuations
- A κ-**rule** for energy allocation
- **Life stages** driven by maturation

Growth patterns (G1–G6)

G1: von Bertalanffy growth

At constant food, most animals follow a decelerating growth curve toward an asymptotic size.

G2–G3: Indeterminate and foetal growth

Many species grow after maturity. Foetal weight increases as the cube of time.

G4–G5: Growth scales with size

Across species: larger species grow more slowly. **Within species:** well-fed individuals grow to a larger size but at a slower rate.

Respiration, feeding, and stoichiometry

Respiration

- Eggs barely respire (mostly reserve)
- Metabolic rate scales with body mass (Kleiber's law, exponent ≈ 0.75)
- Feeding causes a transient increase in metabolism

Feeding

Starving animals survive, grow, and reproduce using stored reserves.

Stoichiometry

Body composition varies with food, but stabilises at constant food — implying at least two pools (structure and reserve).

Three state variables

Structural volume V

Permanent biomass that requires maintenance.

Reserve energy E

Temporary biomass that can be mobilised. No maintenance cost.

Maturity E_H

Cumulative energy invested in complexity. Controls stage transitions (birth, puberty). Not a mass or energy pool.

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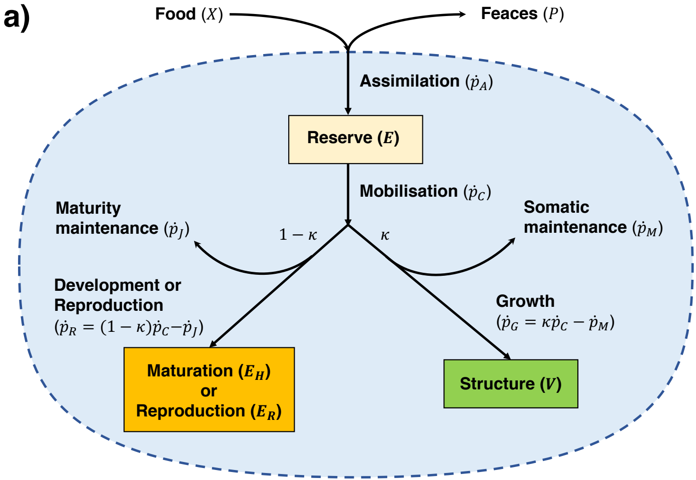
Why two biomass pools?

Reserve decouples **feeding from growth**. An organism can survive, grow, and even reproduce without eating — something a single-pool model cannot explain.

Sousa et al. (2008)

The standard DEB model



Feeding and assimilation

Feeding is proportional to surface area

Maximum ingestion scales with $V^{2/3}$. The actual intake depends on food availability through the functional response $f = X/(X + K)$, which ranges from 0 (no food) to 1 (ad libitum).

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Assimilation

A fixed fraction of the ingested energy enters the reserve:

$$\dot{p}_A = \{\dot{p}_{Am}\}\, V^{2/3}\, f$$

- $\{\dot{p}_{Am}\}$ = surface-specific assimilation rate
- Assimilation efficiency is constant (fixed stoichiometry)

Sousa et al. (2008)

Reserve dynamics

Reserve as a buffer

Energy flows into the reserve (assimilation) and out of it (catabolism). The reserve balance is:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \dot{p}_A - \dot{p}_C$$

where $\dot{p}_A$ is assimilation and $\dot{p}_C$ is catabolism (mobilisation).

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Key properties

- Reserve allows survival during fasting
- Reserve fuels embryonic development (no feeding)
- Reserve buffers environmental fluctuations
- At constant food, reserve density $e = E/V$ stabilises — this is **weak homeostasis**

The κ-rule: energy allocation

How is mobilised energy spent?

Catabolic power $\dot{p}_C$ from the reserve is partitioned at a fixed ratio:

Soma (κ)

- Growth $\dot{p}_G$
- Somatic maintenance $\dot{p}_M$

Maturity/reproduction $(1 - \kappa)$

- Maturation $\dot{p}_R$ (juveniles)
- Reproduction $\dot{p}_R$ (adults)
- Maturity maintenance $\dot{p}_J$

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Consequences

- κ is constant throughout life
- Growth and reproduction do not directly compete (indeterminate growth)
- Once maturity is reached, surplus $(1-\kappa)$ energy goes entirely to reproduction

The DEB growth equation

Combining assimilation, reserve dynamics, and the κ-rule gives the DEB growth equation:

$$\frac{\mathrm{d}L}{\mathrm{d}t} = \frac{\dot{v}}{3}\,\frac{e - L_h/L_m - L/L_m}{g + e}$$

Components

- $e = E/V$ = reserve density (how full the tank is)
- $\dot{v}$ = energy conductance (mobilisation speed)
- $g = [E_G]/(\kappa[E_m])$ = investment ratio (cost of structure relative to reserve capacity)
- $L_m = \kappa\{\dot{p}_{Am}\}/[\dot{p}_M]$ = maximum possible length
- $L_h = \{\dot{p}_T\}/[\dot{p}_M]$ = heating length (surface correction)

Maximum size

What sets the ultimate size?

Growth stops when all available energy is spent on maintenance:

$$L_m = \frac{\kappa\{\dot{p}_{Am}\}}{[\dot{p}_M]}$$

- Higher assimilation $\{\dot{p}_{Am}\}$ $\rightarrow$ larger size
- Higher maintenance $[\dot{p}_M]$ $\rightarrow$ smaller size
- A larger κ (more to soma) $\rightarrow$ larger size

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When food is limited

- At lower food f, ultimate size is smaller: $L_\infty = fL_m - L_h$
- Growth is slower because less energy reaches the reserve
- This is the mechanism behind pattern G5

von Bertalanffy growth from DEB

When food and temperature are constant, reserve density stabilises ($e = f$). The DEB growth equation simplifies to:

$$\frac{\mathrm{d}L}{\mathrm{d}t} = \dot{r}_B(L_\infty - L)$$

Why this matters

- Pütter and von Bertalanffy **assumed** this curve
- DEB **derives** it from first principles
- When food or temperature varies, DEB predicts *departures* from the simple curve
- The parameters $\dot{r}_B$ and L_∞ now have mechanistic interpretations

What DEB explains

Core mechanism

Biomass is partitioned into **structure** (permanent, requires maintenance) and **reserve** (temporary, no maintenance).

Empirical patterns

- von Bertalanffy growth
- Indeterminate growth
- Foetal weight $\propto t^3$
- Growth rate $\propto 1/L_\infty$
- Kleiber's law

Key assumptions

- Feeding $\propto V^{2/3}$
- Fixed stoichiometry
- κ-rule allocation
- Weak and strong homeostasis

Connection to the course

Day 1–2: Biophysical ecology

- Energy and mass **balance** equations
- Body temperature from environment
- NicheMapR implementation

Day 3: DEB theory

- Energy **allocation** within organisms
- Reserve, growth, reproduction
- Life-history consequences

Next: NicheMapR implementation

The next presentation shows how DEB theory is implemented in NicheMapR (Kearney & Porter, 2020), from parameter estimation to coupled biophysical simulations.

The Add-my-Pet (AmP) database

What is AmP?

A global, community-curated collection of DEB parameter estimates for over 3000 animal species.

- Started by Bas Kooijman, now a collaborative online resource
- Each species entry includes parameter values, original data, and references

https://www.bio.vu.nl/thb/deb/deblab/add_my_pet.html

Linking data to DEB parameters

Core principle

DEB parameters cannot be measured directly, but they can be estimated from observable life-history traits such as age and size at birth, age and size at puberty, ultimate size, and maximum reproductive rate.

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The parameter estimation table

Each observable trait maps to a combination of DEB parameters:

- Age at birth $\rightarrow$ maturity threshold E_H^b
- Age at puberty $\rightarrow$ maturity threshold E_H^p
- Length at birth/puberty/adulthood $\rightarrow$ assimilation rate, maintenance rate, κ, and the shape coefficient δ_M
- Maximum reproduction rate $\rightarrow$ reproduction efficiency and energy cost per offspring

The shape coefficient δ_M

Bridging theory and measurement

DEB uses structural length L, but we measure physical length L_w. The shape coefficient converts between them:

$$L = L_w/\delta_M$$

- δ_M is estimated from weight–length data
- A single value is used for a species, constant across food and temperature
- Getting δ_M wrong affects all size-dependent predictions

Kearney (2021)

From AmP to simulation-ready parameters

1. Load `allStat` — a named list of species entries
2. Check the model type for your species (`std`, `abj`, `abp`)
3. Extract the 12 primary DEB parameters and the 4 temperature correction parameters
4. Handle missing fields — not all species have complete data; some fields depend on the model type

Theory $\leftrightarrow$ code name mapping

One concept, many names

The same DEB parameter appears under different names in the literature, AmP, and NicheMapR. Key mappings:

- $\{\dot{p}_{Am}\}$ (assimilation rate) $\rightarrow$ `p_Am` in both AmP and NicheMapR
- $\dot{v}$ (energy conductance) $\rightarrow$ `v`
- κ (allocation fraction) $\rightarrow$ `kap`
- $[\dot{p}_M]$ (maintenance rate) $\rightarrow$ `p_M`
- $[E_G]$ (cost of structure) $\rightarrow$ `E_G`
- E_H^b, E_H^p (maturity thresholds) $\rightarrow$ `E_Hb, E_Hp`
- δ_M (shape coefficient) $\rightarrow$ `del_M`

Four DEB functions for different needs

- `DEB()`: DOPRI5 adaptive solver, most accurate, best for stand-alone simulations of a single individual
- `DEB_const()`: deSolve-based, handles life-cycle events (hatching, birth, puberty) automatically, for constant temperature and food
- `DEB_var()`: Same as `DEB_const()` but accepts variable temperature and food time series
- `DEB_euler()`: Fixed-step Euler method, fastest, used inside the coupled ectotherm model

R-side vs. Fortran-side

The first three functions run in R. `DEB_euler()` runs inside the Fortran biophysical model for speed.

Stand-alone simulation with `rundeb()`

The wrapper function

`rundeb()` provides a single interface to all four DEB solvers:

- You provide: species name, number of days, temperature and food time series
- It runs the appropriate solver and returns a 38-column data frame (`debout`)
- Additional outputs include mass-balance and power-flux summaries

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What `debout` contains

- Stage (egg, juvenile, adult), structural volume, reserve energy, maturity, reproduction buffer
- Derived quantities: length, wet mass, gonad mass, gut content
- Metabolic fluxes: O_2 consumption, CO_2 production, N-waste, water

Choosing the right solver

Decision guide

- **Constant environment, want automatic events?** → `DEB_const()`
- **Variable temperature or food?** → `DEB_var()`
- **Maximum accuracy needed?** → `DEB()` (DOPRI5 adaptive step)
- **Inside the coupled biophysical model?** → `DEB_euler()` (fast, fixed step)

Model type matters

The solver is chosen automatically based on the DEB model type (standard, abj, abp), which controls the number of state variables and the event structure.

Why couple biophysics and DEB?

Biophysics provides:

- Body temperature
- Activity status
- Microclimate conditions

DEB provides:

- Size-dependent surface area
- Metabolic heat production
- Growth, reproduction, survival

The feedback loop

Body temperature and activity control DEB rates. DEB's size and heat output feed back into the energy balance. Only coupled simulations capture this two-way interaction.

Software architecture

1. Meteorological data → `micro_global()` → hourly microclimate
2. Microclimate + DEB parameters + biophysical parameters → `ectotherm(DEB = 1)`
3. `ectotherm()` calls the Fortran biophysical model (`ECTORUN`), which runs hourly
4. At each hour, DEB is called: update size, reserve, maturity, reproduction
5. Updated size feeds back into the biophysical calculations for the next hour

Two time scales

Microclimate is typically daily (interpolated to hourly). The ectotherm model runs hourly. DEB is called every hour. Multi-year runs recycle the same daily microclimate pattern.

Coupling point 1: temperature

T_b controls all DEB rates

The three core rate parameters (assimilation `p_Am`, energy conductance `v`, maintenance `p_M`) are corrected by an Arrhenius function of body temperature. Higher T_b means faster rates — up to thermal limits.

DEB produces metabolic heat

Somatic maintenance, maturity maintenance, and reproduction overhead all generate heat. This heat enters the biophysical energy balance alongside solar and thermal radiation — a feedback that matters for large or active animals.

Coupling point 2: size and shape

Size feeds back to biophysics

As an animal grows, its surface area increases. This changes:

- **Solar radiation** absorption (bigger area)
- **Convection** (boundary layer scales with size)
- **Evaporative** water loss (more skin area)

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A growing individual experiences a changing climate

An animal that doubles in length has four times the surface area. Its heat balance is fundamentally different from that of a juvenile. Only coupled simulations capture this ontogenetic shift.

Coupling point 3: activity and feeding

Behaviour connects environment to DEB

The biophysical model determines activity from body temperature:

- **Active** (foraging): T_b within voluntary range
- **Basking**: T_b below voluntary minimum
- **In shelter**: T_b extreme or night

Feeding in DEB

Only active hours count toward feeding. The functional response f (ranging 0–1) determines how much food is actually assimilated. Assimilation is proportional to both f and surface area.

Coupling point 4: water budget

DEB produces and consumes water

From the DEB mass balance, NicheMapR derives several water fluxes:

- **Metabolic water** from respiration
- **Food water** from ingested prey
- **Faecal water** lost with waste
- **N-waste water** used to excrete nitrogen

Why this matters

These fluxes enter the biophysical water balance alongside evaporation and drinking. In arid environments, the cumulative water balance can determine survival.

Data flow in a coupled simulation

Six output streams

- `environ`: hourly body temperature, behaviour
- `enbal`: hourly energy balance components
- `masbal`: hourly water balance components
- `debout`: hourly DEB state variables (stage, length, mass, reserve, maturity, reproduction)
- `yearout`: annual summaries (max size, number of clutches, fecundity)
- `yearsout`: lifetime summaries (lifespan, generation time, R_0, r_{max})

Interpreting DEB outputs

Hourly time series (`debout`)

- Stage transitions: egg $\rightarrow$ juvenile (birth), juvenile $\rightarrow$ adult (puberty)
- Growth curves: how length and mass change with temperature and food
- Reserve dynamics: build up during feeding, deplete during fasting
- Reproduction: the reproduction buffer accumulates and drops sharply at each clutch

Annual and lifetime summaries

- `yearout`: which years were good or bad for growth and reproduction
- `yearsout`: does the population persist? $r_{\text{max}} < 0$ means a sink population

Key takeaways

1. The **AmP database** provides DEB parameters for over 3000 species
2. **NicheMapR** offers four DEB solvers: from stand-alone exploration to coupled biophysical simulations
3. **Coupling** DEB with biophysics captures feedbacks between size, temperature, activity, and water balance
4. The coupled model predicts emergent life-history patterns (size clines, fecundity, range limits) from first principles

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